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IB Physics Internal Assessment

What is the effect of mass on the damping of a simple pendulum?

1 Introduction

I'm a fan of all things historical, and one of my more recent explorations led me to the Wikipedia page on grandfather clocks. Some minutes into reading it, I grew quite surprised — and not because of any interesting tidbit of history. What most jumped out to me instead was the fact that these clocks have to be calibrated (that is, have energy inputted by mechanical winding) only once or so *a week* for their pendulums to keep visibly swinging — a far cry from the handful of minutes offered by Newton cradles at school or swings in a playground. While likely in some part due to clocks' relatively frictionless gears, this still revealed a rather interesting phenomenon, and one which I continued noticing in parks and school labs later: the heavier the pendulum, the less its swings dropped off over time.

When considering the ubiquity of such 'damped' pendulums, from amusement park rides to car suspensions to even swinging doors, this seemed the perfect subject to explore as a Physics IA. This is especially true because I had trouble answering my own question of the effect of mass on pendulum swings; existing literature on the subject is scant.

2 Background Information

This investigation looks at the *simple pendulum*, which can be generalized as a point mass hanging from a massless, inflexible string of length L. The only forces acting on such a system are gravity and the tension of the string, which can respectively be represented by mg (via Newton's second law) and T.

Here, mg is particularly significant — as gravity acts towards the center of the Earth, this means that if displaced from equilibrium, a component of gravitational force will act on the mass and cause it to accelerate towards equilibrium in the opposite direction of its displacement. This acceleration is given by $a = -\omega^2 x$, where x is the pendulum's displacement from equilibrium and ω its angular frequency $\frac{2\pi}{T} = \frac{2\pi}{2\pi\sqrt{\frac{L}{g}}} = \sqrt{\frac{g}{L}}$.

Such a situation is represented by Figure 1. In it, θ is the angle between the string and the vertical, g the acceleration due to gravity, and $mg\sin\theta$ the accelerating force — tension and the remaining component of gravity are equal and opposite, and so have no effect on the system. The angle θ is formally known as the *angular displacement* of a pendulum, and, when at a maximum, is defined as its *angular amplitude*.

To calculate it, the pendulum's length L and its displacement (the arc x) are used:

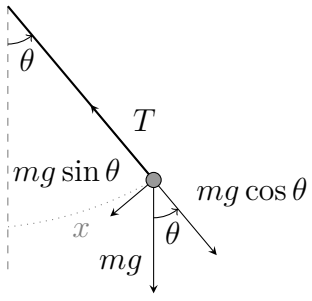


Figure 1: Forces on a simple pendulum

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$$\theta = \frac{x}{L} \tag{1}$$

As shown in the Physics data booklet, this displacement is just the below function of time:

$$x(t) = x_{max} \cos \omega t \tag{2}$$

Since cos is a periodic function, this suggests that the pendulum will oscillate indefinitely. However, in reality, forces such as air resistance between the pendulum and air cause the system to lose energy — and, in turn, the pendulum to continuously slow down and eventually stop[1]. This effect is known as *damping*, and is caused by a damping force F_d acting on the system. This force is proportional to a function of velocity $F(v)$ and acts in its opposite direction, so for small speeds it can be approximated by a first-order Taylor expansion about $v = 0$:

$$F_d = F(v) = -\lambda_1 v - \lambda_2 v^2 + \cdots \approx -\lambda v \tag{3}$$

where λ is a damping constant and v pendulum velocity. When this force is added to the system, the net force becomes Equation 4.

$$\Sigma F = -\omega^2 x - \lambda v = ma \tag{4}$$

Writing this as a differential equation $ma + \lambda v + \omega^2 x = 0 = m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx$ and solving for x as outlined Kharkongor & Mahato[2] for small angles yields the displacement of a damped pendulum as a function of its mass and time:

$$x(t) = x_{max} e^{-\frac{\lambda}{2m}t} \cos(\omega t) \tag{5}$$

Which is a relationship that can be visually represented below.

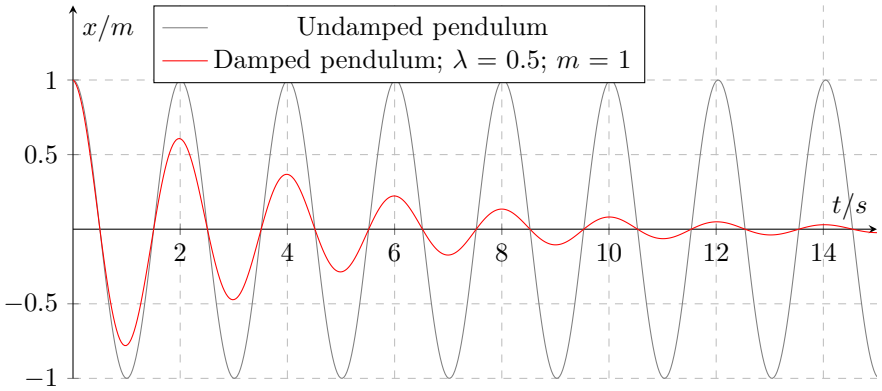


Figure 2: Example graphs of angular displacement over time for a damped and undamped pendulum

[1] Pirooz Mohazzabi and Siva Shankar. "Damping of a Simple Pendulum Due to Drag on Its String". In: *Journal of Applied Mathematics and Physics* 05 (Jan. 2017), pp. 122–130. DOI: 10.4236/jamp.2017.51013.

[2] D Kharkongor and Mangal C Mahato. "Resonance oscillation of a damped driven simple pendulum". In: *European Journal of Physics* 39.6 (2018), p. 065002.

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3 Hypothesis

Equation 5 demonstrates that displacement (and so amplitude, given $\theta = \frac{x}{L}$ from Equation 1) of a damped pendulum decreases exponentially over time. An exact mathematical model for angular amplitude can then be made by substituting in angular frequency ω, which is given by the below for linearly-damped systems and small angles.

$$\omega = \sqrt{\frac{g}{L} - \left(\frac{\lambda}{2m}\right)^2} \tag{6}$$

Since this investigation explores angular amplitude after exactly ten oscillations, the time period for each of these oscillations is $T = 2\pi/\omega$. By then rewriting for angular amplitude using $\theta = \frac{x}{L}$, this condenses to Equation 7:

$$\theta(m) = \frac{x_{max} e^{-\frac{10\pi\lambda/\sqrt{\frac{g}{L}-(\frac{\lambda}{2m})^2}}{m}} \cdot cos\left(\sqrt{\frac{g}{L} - (\frac{\lambda}{2m})^2} \cdot \frac{20\pi}{\sqrt{\frac{g}{L}-(\frac{\lambda}{2m})^2}}\right)}{L} = \frac{x_{max} e^{-\frac{10\pi\lambda/\sqrt{\frac{g}{L}-(\frac{\lambda}{2m})^2}}{m}}}{L} \tag{7}$$

The above may be somewhat inaccurate as its derivation involved the small-angle approximation and assumed strictly linear damping. However, using these is justifiable — Equation 4 would otherwise be analytically insoluble, providing no avenue for exact modeling. In any case, I believe the impact of these simplifications is minimal — after all, the small-angle approximation will quickly become valid regardless of release angle due to damping exponentially decreasing angular amplitude. Furthermore, linear damping is the most common form of damping used in literature on this subject, suggesting that it is an accurate approximation.

As such, I predict that the angular amplitude of a pendulum after 10 full oscillations is related to mass by Equation 7.

4 Experiment Design

4.1 Variables

The **independent variable** for this experiment was the mass of the pendulum bob. Eleven different masses, ranging from 10 g to 110 g in 10 $\pm$ 0.1 g intervals, were placed within a plastic bag and tied to two strings of negligible mass to form a pendulum.

This range was chosen to ensure that the pendulum would not be so light as to have the string and plastic bag make up the majority of the system's weight, but also not so heavy that the angular amplitude of the 10th oscillation be too large to be recorded.

The **dependent variable** was the pendulum's angular amplitude after ten full oscillations.

Multiple **control variables** were used to ensure a fair experiment:

- The vertical length of the pendulum was kept constant at 100 cm ($\pm$ 0.1 cm).
- All trials were released from the same starting angle of 0.989 radians and from rest (initial angular velocity was zero). This large starting angle was necessary to accurately record the angular amplitude of smaller masses after 10 oscillations.

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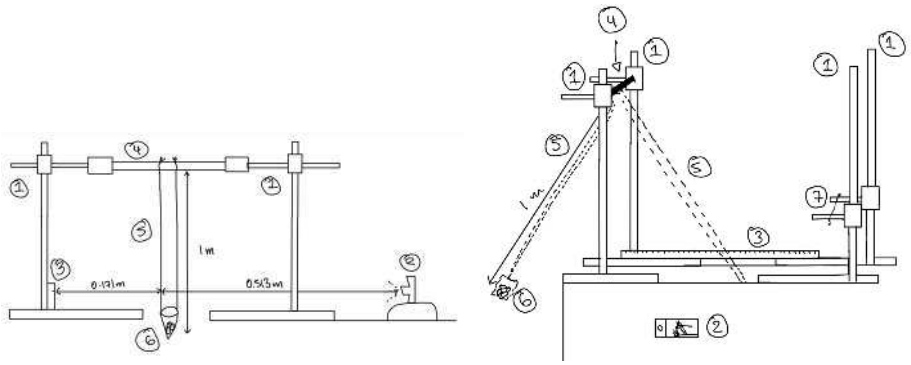
- The strings holding the masses (0.5 g, ± 0.1 g) and bag they were enclosed in (1 g, ± 0.1 g) were kept taut at all times.
- The distances between the camera and second string, and second string and ruler were kept constant at 51.3 cm and 17.1 cm respectively (both ± 0.1 cm). This was done to uniformly mathematically account for variable parallax error.
- All trials were conducted in the same room, and one with a constant room temperature, to keep constant the damping constant λ (as it is influenced by air resistance).
- A set square minimized swinging in the y-direction by ensuring that the two strings holding the mass were parallel to each other and perpendicular to the ground.

4.2 Apparatus

- Reel of twine (1)
- 10 g mass (± 0.1 g) (11)
- High-speed camera (1)
- Set square (1)
- Meter ruler (± 0.1 cm) (1)
- Metal bar (1)
- Small plastic bag (1)
- Roll of adhesive tape (1)
- Laptop (1)
- QuickTime Player software (1)
- Clamp stand (4)
- Scissors (1)

4.3 Diagrams

Head-on and side views for the experimental setup are shown below.



(a) Front view (b) Side view

Figure 3: Diagrams (not to scale) for experiment. ① – clamp stand; ② – camera; ③ – meter ruler; ④ – metal bar; ⑤ – twine for pendulum; ⑥ – mass/es for pendulum; ⑦ – reference string.

4.4 Methodology

I. Set up the apparatus as outlined in Figure 3, taking care to ensure that combined string and plastic bag length is precisely 1.000 m.

II. Note the marking on the ruler perpendicular to the camera lens. Start recording.

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III. Fill the plastic bag tied to the string with a mass of 10 g, then tape it shut. To minimize twisting of the plastic bag as it swings, it must be rotated by 90 degrees and hence cannot be ziplocked.

IV. Pull the pendulum back so that its bottom just touches the reference string shown on the right side of Figure 3b (which leads to an angle of release of 0.989 ± 0.006 radians).

V. Release the pendulum and allow it to oscillate ten full times.

VI. Catch the pendulum and repeat steps III. – IV. twice more.

VII. Repeat steps III. through VI. for remaining masses, in 10 g intervals up to 110 g.

VIII. Stop recording and send video to laptop computer.

IX. Go through recorded video frame-by-frame in QuickTime Player to find the angular amplitude of the pendulum for each trial after 10 oscillations. account for parallax error.

4.5 Risk Assessment

No significant ethical concerns were raised by this experiment. The only potential environmental concern was using a non-biodegradable plastic bag, but since this bag originally held marbles, it was only repurposed temporarily and hence didn't create waste.

However, several safety precautions were taken. Firstly, the experiment was conducted in a corner of the room far from other students to prevent equipment from falling on / the pendulum from hitting anyone. Secondly, the experiment was performed at desk-level height so that even if equipment (which was tightly screwed) did fall, injury sustained would be minimal. Finally, the plastic bag was sealed to prevent masses from flying out and injuring classmates.

5 Results

5.1 Raw Data

		Apparent horizontal displacement in cm after 10 oscillations			
		Trial 1 (± 0.1 cm)	Trial 2 (± 0.1 cm)	Trial 3 (± 0.1 cm)	Average (± 0.1 cm)
Mass in grams (± 0.1 g)	10.0	-5.4	-5.2	-4.9	-5.2
	20.0	0.7	0.5	0.6	0.6
	30.0	5.7	5.9	5.9	5.8
	40.0	10.6	9.9	10.3	10.3
	50.0	13.6	14.1	13.9	13.9
	60.0	17.5	17.6	17.5	17.5
	70.0	20.3	20.5	20.7	20.5
	80.0	23.8	23.7	23.3	23.6
	90.0	26.3	26.3	26.6	26.4
	100.0	28.1	29.3	30.0	29.1
	110.0	32.6	31.0	30.5	31.4

Table 1: Raw data collected from the experiment

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5.2 Processed Data

Mass in grams (± 0.1 g)	Horizontal displacement after 10 oscillations (all ± 0.1 cm) Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average (cm)	Avg. final θ (rads)
10.0	2.8	3.0	3.2	3.0	0.053 ± 0.002
20.0	7.4	7.3	7.3	7.3	0.13 ± 0.00
30.0	11.2	11.3	11.3	11.3	0.195 ± 0.002
40.0	14.8	14.3	14.6	14.6	0.252 ± 0.002
50.0	17.1	17.5	17.3	17.3	0.296 ± 0.002
60.0	20.0	20.1	20.0	20.0	0.339 ± 0.002
70.0	22.1	22.3	22.4	22.3	0.374 ± 0.002
80.0	24.7	24.7	24.4	24.6	0.409 ± 0.002
90.0	26.6	26.6	26.8	26.7	0.439 ± 0.001
100.0	28.0	28.9	29.4	28.8	0.469 ± 0.001
110.0	31.3	30.1	29.8	30.4	0.491 ± 0.001

Table 2: Experimental data processed for parallax and used to derive θ

5.2.1 Parallax Calculation

In this experiment, the camera was not on perpendicular to the y-axis of the pendulum's rest position, but rather 27.5 (± 0.1) cm to the right of it in order to have a greater field of view. This led to parallax error, illustrated through the negative displacement measurements in the first row of Table 1. The resolution to this error is shown through Figure 4 below.

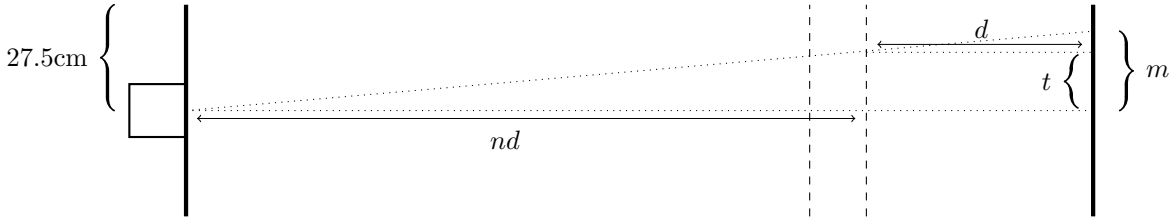


Figure 4: Experimental parallax error and its accounting-for

In the diagram, the horizontal displacement m as measured by the camera is greater than its true displacement from the camera t. The difference between these two, $m - t$, is the parallax error. This can be eliminated using the principle of similar triangles. The first triangle is formed by $nd + d$, m, and the full diagonal; it represents the measured value. The second triangle has edges nd, t, and the left part of the diagonal. Being similar, the ratio of these two triangles ($\frac{nd+d}{nd}$) can be used to give an equation for t alone:

$$t = \frac{nd}{nd + d} \cdot m \tag{8}$$

As explained in Section 4.1, nd is 51.3 cm and d 17.1 cm. Such an arrangement creates a neat n of 3, and so, for example, gives the below as the true average horizontal displacement from the camera for a 20 g pendulum after ten oscillations, using raw data from Table 1:

$$t = \frac{3 \cdot 17.1}{3 \cdot 17.1 + 17.1} \cdot (27.5 - 0.6) = 20.2 \text{ cm (3 s.f.)} \tag{9}$$

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Note that t is the true displacement of the pendulum's horizontal position *from the camera*. Since the camera is perpendicular to the 27.5 cm ruler mark, t must be then subtracted from 27.5 to get the true displacement from the zero-mark on the ruler (the pendulum rest position). Doing so here yields the correct Table 2 value of $27.5 - 20.2 = 7.3$ cm.

5.2.2 Angular Amplitude Error Calculation

The effect of mass on angular amplitude is the subject of this investigation, so various angular amplitudes and their errors must be calculated in order to effectively evaluate results and confirm their accuracies. Finding the first is a straightforward application of the $\tan\theta \equiv \frac{opposite}{adjacent}$ identity. Here, the *opposite* and *adjacent* sides are respectively the horizontal and vertical displacements, shown in Figure 5 as d_h and d_v.

Angular amplitude is therefore:

$$\theta = \arctan\left(\frac{d_h}{d_v}\right) \tag{10}$$

However, calculating the error is a more involved process. Since arctan is not a linear function, the error cannot be calculated by simply adding or multiplying absolute errors in d_v and d_h. Instead, it must be estimated using the general formula for error propagation, shown below:

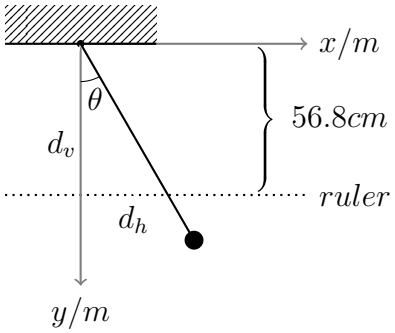


Figure 5: Visualization of the trigonometric calculation of θ

$$s_f = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 s_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 s_y^2 + \left(\frac{\partial f}{\partial z}\right)^2 s_z^2 + \cdots} \tag{11}$$

In the context of the two-variable arctan function, this becomes Equation 12, whose partial derivatives of θ with respect to d_h and d_v are respectively Equations 13 and 14.

$$\Delta\theta \approx \sqrt{\left(\frac{\partial\theta}{\partial d}\Delta d_h\right)^2 + \left(\frac{\partial\theta}{\partial d_v}\Delta d_v\right)^2} \tag{12}$$

$$\frac{\partial\theta}{\partial d_h} = \frac{1}{1+\left(\frac{d_h}{d_v}\right)^2}\cdot\frac{1}{d_v} \tag{13}$$

$$\frac{\partial\theta}{\partial d_v} = \frac{1}{1+\left(\frac{d_h}{d_v}\right)^2}\cdot\frac{-d_h}{d_v^2} \tag{14}$$

The above error measure is an estimate valid only when $\Delta d_h \ll d_h$ and $\Delta d_v \ll d_v$[3]. This holds true for this experiment, as the uncertainty in the meter ruler used to measure vertical displacement was negligible in comparison to the value measured, being $\pm$ 0.1 cm of 56.8

[3]Cornell. *Propagation of Errors*. `https://courses.cit.cornell.edu/virtual_lab/LabZero/Propagation_of_Error.shtml`. (Accessed on 10/12/2022).

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cm. The horizontal displacement uncertainty was likewise ± 0.1 cm, so also much smaller than its measured values between 3.0 and 30.4 cm.

In any case, finishing the process by substituting sample values of $d_h = 0.568$, $\Delta d_h = 0.001$, $d_v = 0.304$, and $\Delta d_v = 0.001$ from the last row of Table 2 gives the correct angular amplitude uncertainty of 0.002 radians.

$$\begin{aligned}\Delta\theta &\approx \sqrt{\left(\frac{1}{0.304(\frac{0.568^2}{0.304^2}+1)}\cdot 0.001\right)^2 + \left(-\frac{0.568}{0.304^2+0.568^2}\cdot 0.001\right)^2} \\ &\approx 0.0015522262 \approx 2\times 10^{-3}\end{aligned} \tag{15}$$

This uncertainty was calculated to be the same for every value of d_h and d_v used in the experiment (as demonstrated by Table 2). As such, it is important in that it demonstrates that all uncertainties in experimental results can be taken to be negligible.

6 Discussion

Processed experimental results are visually represented in Figure 6. No error bars are used for the angular amplitude after 10 oscillations, as the percentage uncertainty (see Table 2) is too small to be meaningfully added, having a maximum value of 0.37%. Similarly, because the pendulum's motion was tracked using a 60 frames-per-second camera, uncertainty beyond the scale of the ruler is likewise neglible. This is because the pendulum's speed decreases to zero. The pendulum is hence 'slow enough' for 'long enough' to have its amplitude accurately photographed to within the ruler's 0.1 cm uncertainty.

However, error bars *are* used in Figure 6 for the angular amplitude of pendulum release, as this had a somewhat significant ∼1 cm uncertainty. Experimental method aimed to minimize this by releasing the plastic bag with the pendulum masses from the top of a fixed reference string, but the bag's contents were not perfectly distributed each time, so starting horizontal displacement and hence calculated angle of release varied slightly.

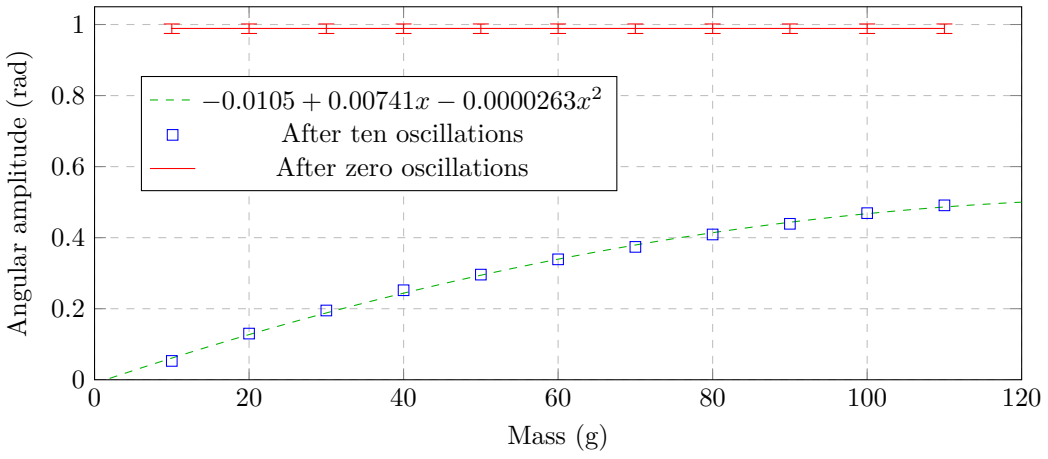


Figure 6: Average angular amplitude after zero and ten oscillations of weighted pendulum

Preliminary results suggest that a pendulum's damping depends on a quadratic and linear

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function of its mass. I found this relationship to be mathematically $-0.0105 + 0.00741x - 0.0000263x^2$, a model shown in green in Figure 6. This model appears to be very accurate, as it fits all data points to within 0.01 radians. While this error is greater than the maximum 0.37% percentage uncertainty in my measured data, it is still comparatively small, and so may simply the result of minor systematic errors discussed later.

That said, despite its close fit, this quadratic model can at best only be a good approximation for smaller pendulum masses. This is because it has two positive roots, $x = 1.424$ and $x = 280.325$ — which, in the context of the negative x^2 coefficient, would suggest that the angular amplitude of a pendulum after ten full oscillations is negative for masses greater than .280325 kg. This cannot be the case; a complete pendulum oscillation always ends on the same side of its equilibrium point as it started, and so, final measured angular amplitude must be positive for all masses in the reference frame of this experiment.

Therefore, it may prove fruitful to compare results directly with my prediction (reproduced below for convenience):

$$\theta(m) = \frac{x_{max} e^{-\frac{10\pi\lambda/\sqrt{\frac{g}{L}-(\frac{\lambda}{2m})^2}}{m}}}{L} \tag{16}$$

In Equation 16, g, x_{max}, and L are constants as outlined earlier, and m is the independent variable. Using the arc length equation $x = r\theta$ with aforementioned initial conditions $r = L = 1.000\,m$ and $\theta = 0.989$ radians, the value of x_{max} is 0.989 m. This leaves only the damping constant λ to be found. By solving computationally for the average measured 10th-oscillation angular amplitude of each experimental mass (with a custom Python script I wrote), the following values of λ are generated:

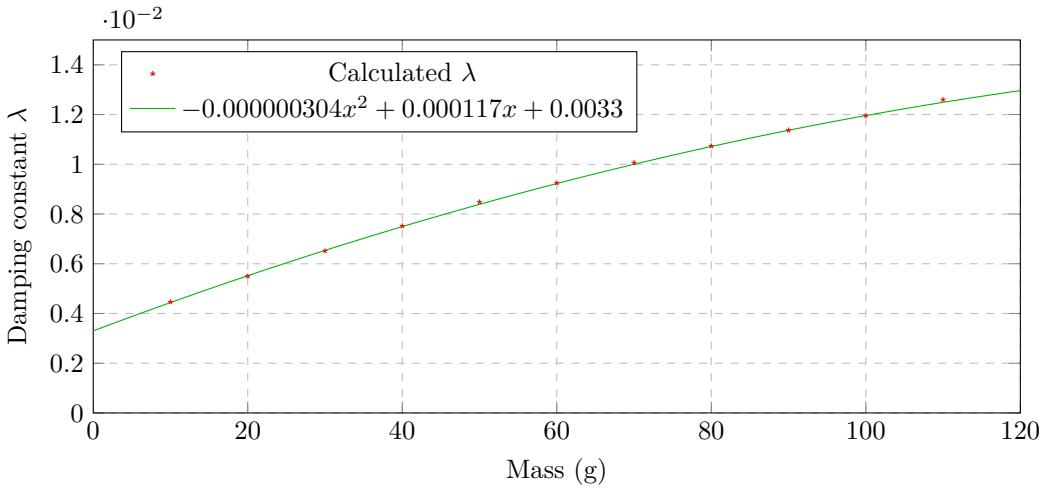


Figure 7: Damping constant calculated using individual data points

If the predicted angular amplitude after ten oscillations for various masses (Equation 16) is then plotted using the average, minimum, and maximum values of λ, Figure 8 (see next page) is produced:

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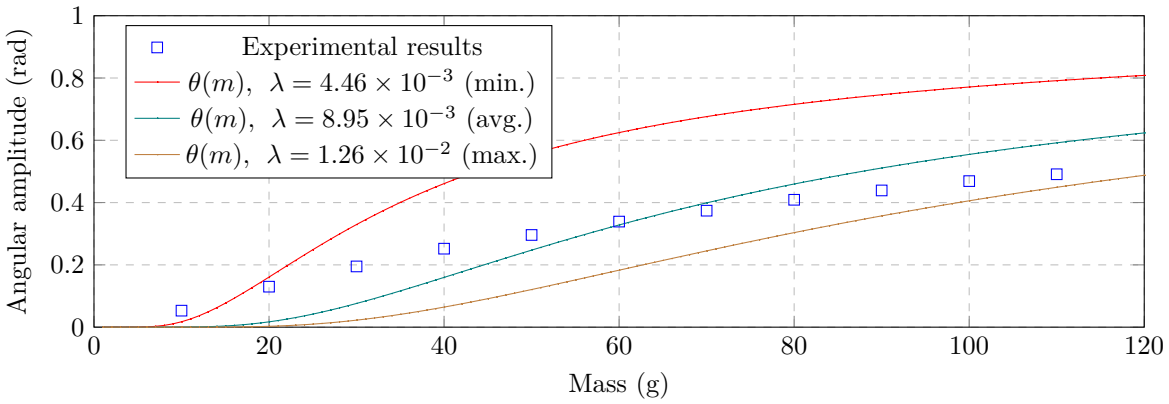


Figure 8: Predicted vs experimental angular amplitudes for various masses after 10 oscillations

Comparing the average damping constant to experimental data gives the best match, the teal line. The quality of this match can be found by calculating its *residual standard error (RSE)*, which is the average absolute difference between expected and predicted values — a measure valid for non-linear models such as this one. Subsituting the average $\lambda = 8.95 \times 10^{-3}$ value into my hypothesis, RSE is 0.0669 (3 s.f.) radians. Considering the 0 to 0.989 radian range of possible amplitudes and comparing with the < 0.001 radian accuracy (RSE) of the quadratic model, this makes the 'average λ model' a relatively poor fit. In fact, even the theoretically most optimal λ is very inaccurate — with another Python script, I found that the best RSE is the still-comparatively-large 0.0655 for a λ of 9.46×10^{-3}.

The poor quality of this model is made especially clear by the fact that my calculated values for λ are not consistent. As shown in Figure 7, λ spans from 4.46×10^{-3} to 1.26×10^{-3} (182.5% variation), and appears to be quadratically dependent on mass, with a line of best fit of $y = -0.000000304x^2 + 0.000117x + 0.0033$. However, λ is defined as the damping *constant* — for a fixed pressure and surface area (which were maintained as control variables during this investigation) it has a constant value. Since my calculated λ is *not* constant, and my model has a large RSE as shown in Figure 8 by the vertical gap between experimental and theoretical results (most significantly at extremities of the data range, suggesting even larger RSE should greater masses have been used), I conclude that my prediction is not a good model for this data and real-world pendulum motion in general. Outside of random and systematic errors, there are two possible explanations for this.

Firstly, the small angle approximation $\sin\theta \approx \theta$ was used throughout the experiment — most notably during the solving of $ma + \lambda v + \omega^2 x = 0$ to derive an exact solution for the displacement of a damped pendulum. Were it not used, the solution would be an infinite expansion and so I would be unable to generate an exact model for $\theta(m)$. That said, this also meant that later rearranging for θ was inaccurate — and potentially quite so, as this experiment released the pendulum from a relatively large angle of 0.989 radians where the approximation becomes inaccurate. Doing so, however, was necessary in order to reliably measure the angular amplitude of the pendulum after 10 swings, especially for smaller masses. I initially assumed that any effects of this would be minimal, as the pendulum would quickly be damped into a small angle — but this seems to have been incorrect and a major contributor to the invalidity of my prediction.

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Additionally, it's possible that the discrepancy stems from the damping force being quadratic, not linear. A fundamental assumption made about this force was that it is in the strictly linear form $-\lambda v$. However, several studies suggest[4] that a primarily quadratic (or combined linear and quadratic) relationship between pendulum velocity and drag is more realistic, as it takes into account turbulent air flows created. Modeling this is, however, once more analytically impossible and so was not attempted.

7 Conclusion

In conclusion, empirical results suggest that my prediction of mass affecting the 10th angular amplitude of a damped pendulum by Equation 17 is incorrect.

$$\theta(m) = \frac{x_{max} e^{-\frac{10\pi\lambda/\sqrt{\frac{9.81}{L}-(\frac{\lambda}{2m})^2}}{m}}}{L} \tag{17}$$

Instead, for small masses, angular amplitude after ten oscillations is best predicted by a simple quadratic model, Equation 18.

$$\theta(m) = -0.0105 + 0.00741m - 0.0000263m^2 \tag{18}$$

However, the accuracy of this quadratic function drops off dramatically with increasing masses, and it provides no direct insights into the angular amplitudes of pendulums swinging for a variable amount of time. That said, this investigation has been moderately successful — I can still extrapolate to answer my research question with the following:

For small masses, there is a quadratic relationship between the mass and angular amplitude after n oscillations of a damped pendulum. For 10 oscillations in particular, it is given by Equation 18.

8 Evaluation

8.1 Random and Systematic Errors

Error	**Size/Impact**	**Possible Solution**
Oscillations were not fully confined to the x-dimension	For smaller masses, the flow of air over the plastic bag caused (sometimes rather significant) oscillations in the y-dimension which were not accounted for.	Use a rigid, x-, y-, and z-dimension symmetrical container to hold the masses.
Angular amplitude at displacement varied	Pendulum was released by hand by looking at a reference string, which led to a combined parallax error and uncertainty of about 1 cm — a small but impactful error.	Use a more precise mechanism such as a motorized clamp. Alternatively, increase initial angle and/or pendulum length.

Table 3: Breakdown of random experimental errors

[4] Temple Fay. "Quadratic damping". In: *International Journal of Mathematical Education in Science and Technology* 43 (Sept. 2012), pp. 789–803. DOI: 10.1080/0020739X.2011.622806.

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Error	Size/Impact	Possible Solution
Masses of string and plastic bag were not counted	Decreasing impact with growing masses, but quite significant for the smallest ones (ex. ∼20% uncertainty for 10 g).	Use different string lengths for each mass so that the center of mass is always at 1 m.
Plastic bag slightly stretched with increasing weight	The plastic material chosen was relatively rigid, making the maximum stretch less than 1 cm and so impact relatively negligible.	Use a less flexible material as a container, such as a cardboard box.
Twine used created unaccounted-for friction	Friction adds a damping term to the equation of angular amplitude, leading to a large cumulative effect after 10 oscillations for larger masses, and so may also explain theory and experiment discrepancy.	Use a wooden post instead of a metal bar, and hammer the pendulum string into it to reduce kinetic friction. Alternatively, use a metal wire or lower-friction string instead of twine.

Table 4: Breakdown of systematic experimental errors

8.2 Limitations, Accuracy, and Future Work

One major limitation during this experiment was the range of independent variables — only eleven distinct masses were used, all less than 150 g, as a result of a small container size. This made it difficult to draw definitive conclusions about the relationship between mass and damped angular amplitude, since the effect of larger masses and hence general trendline could not be fully confirmed. Repeating the experiment for a greater range of masses is hence a productive avenue for potential future research. It would have also been beneficial to use a solid, inelastic container of fixed volume to hold the masses — using a plastic bag as a control variable made results somewhat less reliable by introducing random spin and slightly increasing the effective length of the pendulum for some masses.

Nevertheless, the majority of outlined errors and limitations were not so significant as to seriously affect the precision of results, as evidenced by the fact that the difference in horizontal displacement across individual columns of processed data was relatively small. This in turn corroborates my conclusion of mass affecting pendulum damping in a quadratic way (at least for small masses and after ten oscillations). However, to make this conclusion more reliable, future extensions to the investigation such as using a longer pendulum and so smaller initial angular displacement or assuming quadratic damping and then solving numerically should be considered.